Mining Media Data I
Winter Semester 26/27
Lecture-2
Affinity Mining I

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Agenda

- Recap: organization and last lecture
- Motivation: what is bought together?
- Itemsets and support
- The search problem
- The Apriori algorithm
- Walk-through on our example
- Outlook: association rules

Recap (1): Website and Assignments

- **Website** (slides, assignments, news): `https://appliedmachinelearning-lab.github.io/teaching/mmdi2627/`
- **Password**: IlyaSutskever
- **Assignments**: presented and discussed on Wednesdays, 14:00–15:30
  - Lecture hall 1.002 (lecture hall X, main building)

Assignment	**Deadline**	**Presentation**
Assignment 1	17 Nov 2026, 23:59 (CET)	Wed, 18 Nov 2026
Assignment 2	15 Dec 2026, 23:59 (CET)	Wed, 16 Dec 2026
Assignment 3	19 Jan 2027, 23:59 (CET)	Wed, 20 Jan 2027

Recap (2): Groups and Registration

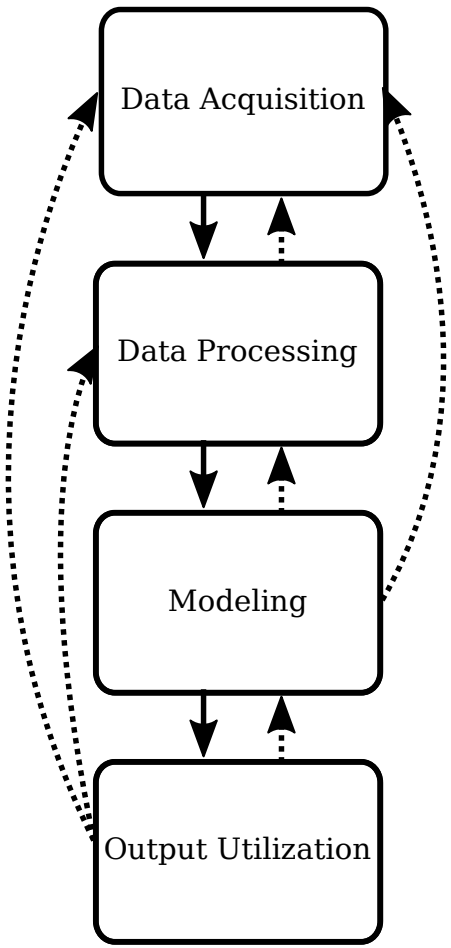
- **Groups**: up to 5 students
- **Group registration** required before the first submission
  - Email to `mchopra@uni-bonn.de` with **group name**
  - For **each** member: first name, last name, matriculation number, Uni-ID
- **Submission**: to `mchopra@uni-bonn.de`, subject: group name and assignment number

Example

Subject: MMDI group registration: The Apriori Avengers

First name	**Last name**	**Matriculation no.**	**Mail**
Ada	Lovelace	1234567	s0alovel@uni-bonn.de
Alan	Turing	7654321	s0aturin@uni-bonn.de

Recap (3): Data Mining Workflow and Tasks



- **Last lecture**: data mining workflow and five tasks
  - Affinity mining, outlier detection, clustering, classification, regression
- **Today**: the first task, **Affinity Mining**
  - Which items occur together?

Customers who bought the item(s):





Also bought :



Affinity Mining: Motivation

- **Goal**: find *interesting* patterns among items in large data
- Typical questions:
  - **Interest analysis**: which items sell together at Christmas?
  - **Content design**: which dungeons came before quitting?
  - **Marketing**: which items to bundle at a discount?
  - **Recommendations**: *customers also bought* ...
- **Interestingness** can mean
  - frequency $\rightarrow$ today
  - conditional probability (confidence) $\rightarrow$ Lecture 3
  - utility, e.g. revenue $\rightarrow$ Lecture 3

Affinity Mining: Real-World Applications

Domain	Items	Transaction	Use
Retail	products	shopping basket	product placement, bundles
E-commerce	products	order history	*frequently bought together*
Streaming	movies, songs	watch list, playlist	recommendations
Healthcare	diagnoses, drugs	patient record	co-occurring diseases
Finance	payment features	payment	fraud patterns

- Same algorithm, different meaning of items and transactions

Running Example: A Game Shop

- In-app shop of a fantasy game
- 6 items, 8 transactions (shopping baskets)
- **Question**: which items are bought together?
- Use: discounts for item pairs, shop design



Elixir Gem Giant Wand Shield Sword Wand

Transaction	**Bought items**
T_1	Elixir, Shield
T_2	Gem, Shield
T_3	Elixir, Sword
T_4	Elixir, Shield, Wand
T_5	Elixir, Sword
T_6	Shield, Wand
T_7	Elixir, Shield, Sword, Wand
T_8	Elixir, Giant Wand

Items, Transactions and Itemsets

- **Items**: a fixed set $A = \{a_1, \ldots, a_m\}$
  - Example: $A = \{\text{Elixir}, \text{Gem}, \text{Giant Wand}, \text{Shield}, \text{Sword}, \text{Wand}\}$, $m = 6$
- **Transaction**: the items of one observation, $T_i \subseteq A$
  - Example: $T_4 = \{\text{Elixir}, \text{Shield}, \text{Wand}\}$
- **Database**: all transactions, $D = (T_1, \ldots, T_n)$
  - Example: $n = 8$ transactions
- **Itemset**: any subset $B \subseteq A$; with $|B| = k$ it is a **k-itemset**
  - Example: $\{\text{Shield}, \text{Wand}\}$ is a 2-itemset

Support

- **Support**: fraction of transactions that contain B

$$\text{supp}(B) = \frac{1}{n} \left| \{T_i \in D \mid B \subseteq T_i\} \right|$$

- Examples from the game shop:

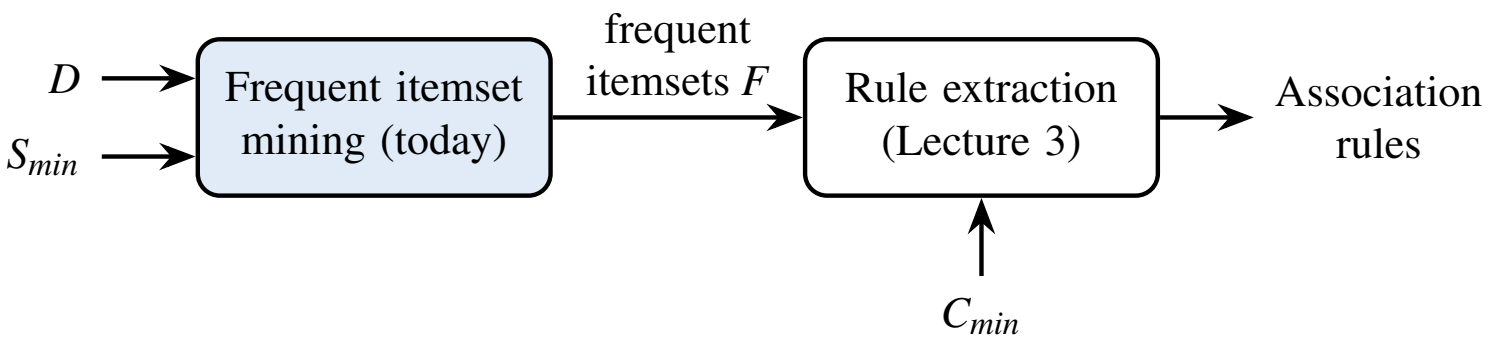
Itemset B	**Contained in**	$\text{supp}(B)$
$\{\text{Elixir}\}$	$T_1, T_3, T_4, T_5, T_7, T_8$	$6/8 = 0.75$
$\{\text{Shield}, \text{Wand}\}$	T_4, T_6, T_7	$3/8 = 0.375$
$\{\text{Elixir}, \text{Shield}, \text{Wand}\}$	T_4, T_7	$2/8 = 0.25$

Frequent Itemset Mining

- B is **frequent** if $\text{supp}(B) \geq S_{min}$ for a threshold $S_{min} \in (0, 1]$
- **Goal**: find *all* frequent itemsets

$$F = \{B \subseteq A \mid \text{supp}(B) \geq S_{min}\}$$

- Example: $S_{min} = 0.25$, i.e. contained in at least 2 of 8 transactions



The Search Problem

- **Brute force**: compute the support of every itemset
- m items $\rightarrow 2^m - 1$ nonempty itemsets
- Each support needs a scan over all n transactions
- The search space grows **exponentially** in m

$\Rightarrow$ We need to **prune** the search space

Items m	**Itemsets** $2^m - 1$
6	63
20	$\approx 10^6$
100	$\approx 10^{30}$
10 000	$\approx 10^{3010}$

Game shop: $m = 6$
Online shop: $m \gg 10\,000$

The Apriori Principle

Apriori principle (downward closure)

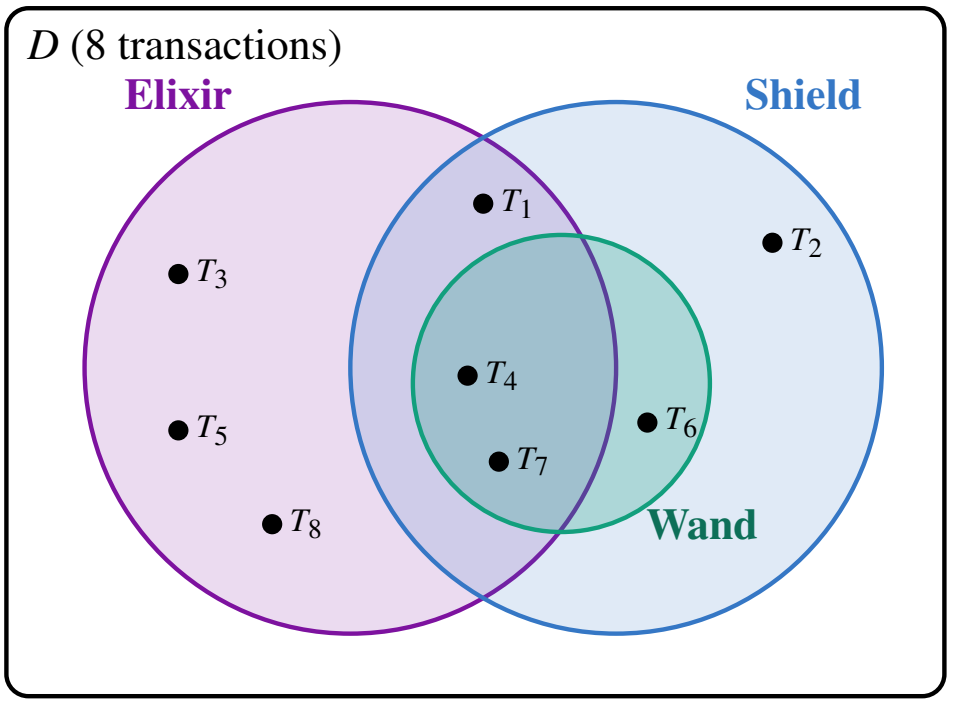
Every nonempty subset of a frequent itemset is also frequent.

- **Why?** Every transaction that contains $B \cup \{a\}$ also contains B:

$$\text{supp}(B \cup \{a\}) \leq \text{supp}(B)$$

- **Turned around**: if B is infrequent, every superset of B is infrequent
- **Example**: $\text{supp}(\{\text{Shield}, \text{Sword}\}) = 0.125 < S_{min}$
  $\Rightarrow$ $\{\text{Elixir}, \text{Shield}, \text{Sword}\}$ cannot be frequent, no need to count it

The Apriori Principle: Visualized



- Dot = transaction, circle = transactions containing the item
- **Support** = share of dots in a region
- **More items** = intersection of circles = **smaller region**

$$\begin{aligned} \text{supp}(\{\text{Shield}\}) &= 5/8 \\ \geq \text{supp}(\{\text{Shield}, \text{Wand}\}) &= 3/8 \\ \geq \text{supp}(\{\text{Elixir}, \text{Shield}, \text{Wand}\}) &= 2/8 \end{aligned}$$

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The Apriori Algorithm

- Level-wise search: from k-itemsets to $(k+1)$-itemsets
- Three steps per level: **generate**, **prune**, **count**
- C_k: **candidate** k-itemsets, support not yet counted
- F_k: **frequent** k-itemsets, support counted and $\geq S_{min}$

Procedure `Apriori(`D, S_{min}`):`

- $F_1 \leftarrow$ all frequent 1-itemsets
- $k \leftarrow 1$
- **while** $F_k \neq \emptyset$ **do**
  - **generate**: $C_{k+1} \leftarrow \{A \cup B \mid A, B \in F_k,\ |A \cup B| = k+1\}$
  - **prune**: remove all $B \in C_{k+1}$ with an infrequent k-subset
  - **count**: $F_{k+1} \leftarrow \{B \in C_{k+1} \mid \text{supp}(B) \geq S_{min}\}$
  - $k \leftarrow k+1$
- **return** $F_1 \cup \cdots \cup F_k$

The Apriori Algorithm: Generate and Prune

- **Generate (join)**: unite two frequent k-itemsets that differ in exactly one item
  - $\{\text{Elixir}, \text{Shield}\} \cup \{\text{Elixir}, \text{Wand}\} = \{\text{Elixir}, \text{Shield}, \text{Wand}\}$
  - For $k = 1$: all pairs of frequent items
  - The same candidate can result from several pairs: keep it once
- **Prune**: keep a candidate only if *all* its k-subsets are frequent
  - No database access needed
- **Count**: one scan of D for the remaining candidates

Walk-through (1): Level $k = 1$

- $S_{min} = 0.25$
- Count all single items
- Gem and Giant Wand are bought only once: infrequent

$\Rightarrow$ $F_1 =$ $\{\text{Elixir}, \text{Shield}, \text{Sword}, \text{Wand}\}$

Candidate	supp
{Elixir}	0.750
{Gem}	0.125
{Giant Wand}	0.125
{Shield}	0.625
{Sword}	0.375
{Wand}	0.375

Walk-through (2): Level $k = 2$

- Generate: all pairs of items in F_1 (6 candidates)
- Prune: nothing to prune, all 1-subsets are in F_1
- Count: two pairs are infrequent

$\Rightarrow$ F_2 contains 4 itemsets

Candidate	supp
{Elixir, Shield}	0.375
{Elixir, Sword}	0.375
{Elixir, Wand}	0.250
{Shield, Sword}	0.125
{Shield, Wand}	0.375
{Sword, Wand}	0.125

Walk-through (3): Level $k = 3$

- Generate: unite pairs in F_2 whose union has 3 items
- Prune: check all 2-subsets against F_2

Union	**Candidate**	**2-subsets in F_2?**
{Elixir, Shield} ∪ {Elixir, Sword}	{Elixir, Shield, Sword}	no: {Shield, Sword}
{Elixir, Shield} ∪ {Elixir, Wand}	{Elixir, Shield, Wand}	yes
{Elixir, Shield} ∪ {Shield, Wand}	*same*	
{Elixir, Wand} ∪ {Shield, Wand}	*same*	
{Elixir, Sword} ∪ {Elixir, Wand}	{Elixir, Sword, Wand}	no: {Sword, Wand}

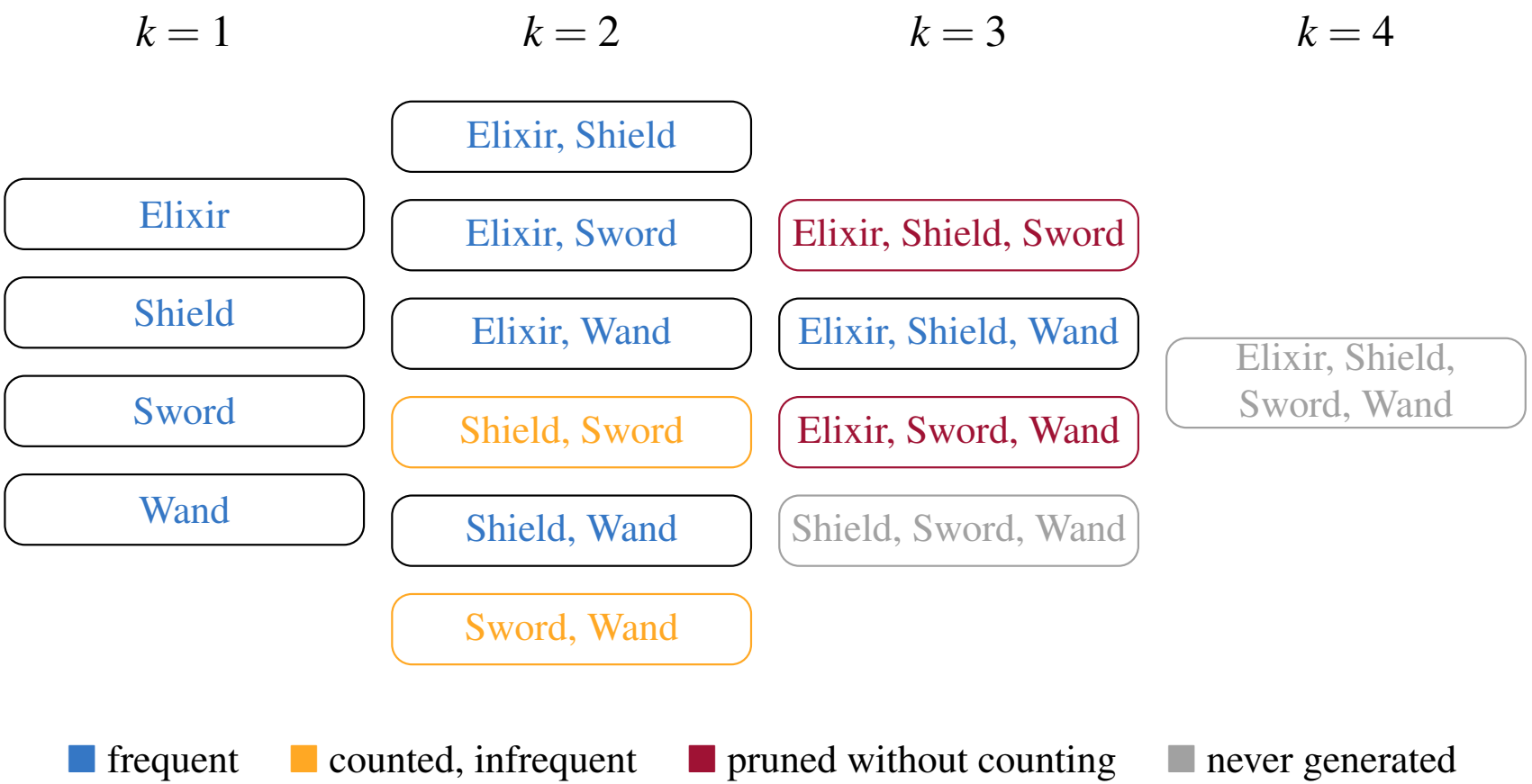
- Count: only one candidate left,
  $\text{supp}(\{\text{Elixir}, \text{Shield}, \text{Wand}\}) = 0.25 \Rightarrow$ frequent
- Level $k = 4$: F_3 has only one itemset, nothing to join $\Rightarrow$ **stop**

Walk-through (4): Result

Frequent itemset	supp
{Elixir}	0.750
{Shield}	0.625
{Sword}	0.375
{Wand}	0.375
{Elixir, Shield}	0.375
{Elixir, Sword}	0.375
{Shield, Wand}	0.375
{Elixir, Wand}	0.250
{Elixir, Shield, Wand}	0.250

- 9 frequent itemsets for $S_{min} = 0.25$
- Supports counted:
  - Apriori: $6+6+1 = \mathbf{13}$
  - Brute force: $2^6 - 1 = \mathbf{63}$
- 3 database scans (one per level)

Walk-through (5): A Look at the Search Space



Outlook: From Frequent Itemsets to Association Rules

- **Association rule** $X \to Y$ with $X \cap Y = \emptyset$: *who buys X also buys Y*
- **Confidence**: empirical conditional probability $p(Y \mid X)$

$$\text{conf}(X \to Y) = \frac{\text{supp}(X \cup Y)}{\text{supp}(X)}$$

- **Strong rule**: $\text{supp}(X \cup Y) \geq S_{min}$ and $\text{conf}(X \to Y) \geq C_{min}$

Rule	conf
Wand $\to$ Shield	1.00
Sword $\to$ Elixir	1.00
Elixir, Wand $\to$ Shield	1.00
Elixir $\to$ Shield	0.50

Details and rule extraction: Lecture 3

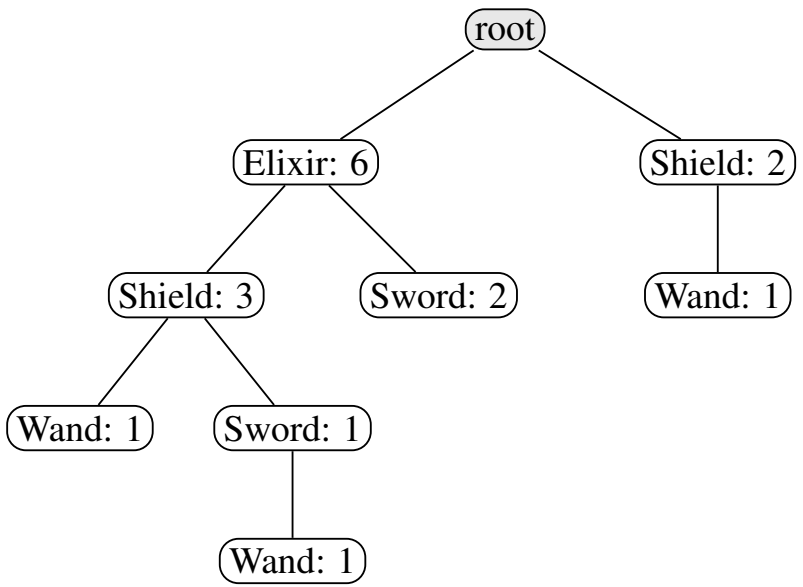
Limitations and Alternatives

- **Limitations of Apriori**
  - Many candidates for large m or small S_{min}
  - One database scan per level
- **Alternative: FP-Growth** (Frequent Pattern Growth)
  - Compresses D into a prefix tree with counts (FP-tree)
  - Mines the tree without candidate generation
  - Only two database scans

FP-Growth (1): Building the FP-Tree

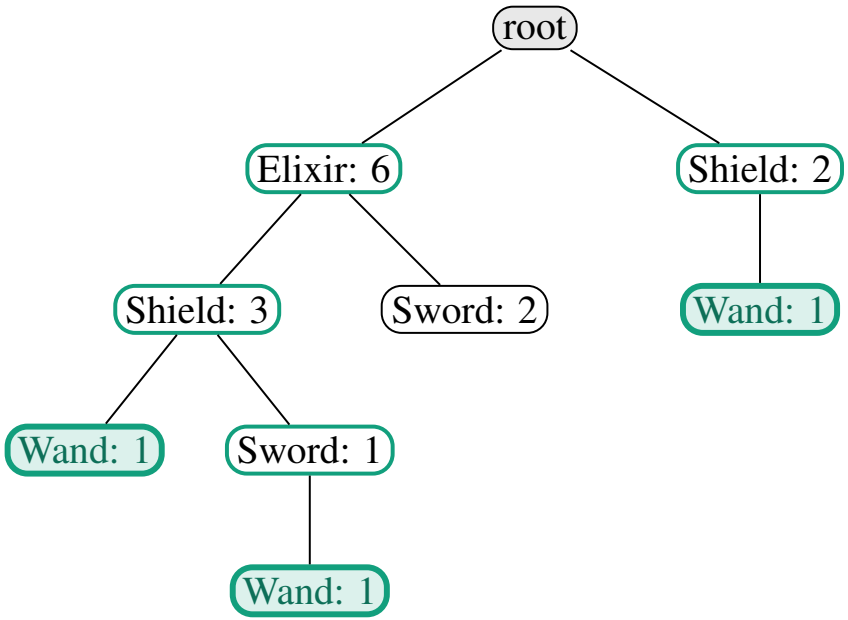
- **Scan 1**: count items, drop infrequent ones, sort by support
- **Scan 2**: insert each transaction as a path from the root
- Shared prefixes are merged, their counts add up
- Sorting by support puts frequent items near the root and maximizes sharing

	Sorted frequent items
T_1	Elixir, Shield
T_2	Shield
T_3	Elixir, Sword
T_4	Elixir, Shield, Wand
T_5	Elixir, Sword
T_6	Shield, Wand
T_7	Elixir, Shield, Sword, Wand
T_8	Elixir



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FP-Growth (2): Mining the Tree



- Example: frequent itemsets with **Wand**
- Paths to the Wand nodes:
  - Elixir, Shield (1)
  - Elixir, Shield, Sword (1)
  - Shield (1)
- Counts: Shield 3, Elixir 2, Sword 1

$\Rightarrow$ {Shield, Wand}: 3/8
{Elixir, Wand}: 2/8
{Elixir, Shield, Wand}: 2/8

- Same result as Apriori, **no candidates**

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Summary

- **Affinity mining**: which items occur together?
- **Support**: fraction of transactions containing an itemset
- **Frequent itemset mining**: find all itemsets with supp $\geq S_{min}$
- **Problem**: $2^m - 1$ candidate itemsets
- **Apriori principle**: subsets of frequent itemsets are frequent
- **Apriori algorithm**: level-wise generate, prune, count

Next Steps

- **Assignment 1**: deadline 17 Nov 2026, 23:59 (CET)
- **Next lecture** (04 Nov): Affinity Mining II, association rules and high-utility itemset mining